

Attachment 2 – Statement of Work

DEFENSE LOGISTICS AGENCY

NICKEL NIOBIUM ALLOY

STATEMENT OF WORK

I. INTRODUCTION

A. BACKGROUND

The Department of Defense (DoD) procures parts that contain niobium. Niobium is commonly used in many forms; however, this statement of work (SOW) is only for nickel niobium alloy, which is used as a master alloy for enhanced strength in steel production.

B. OBJECTIVES

The objective of this procurement is to establish Indefinite Delivery Indefinite Quantity (IDIQ) requirement type contract ordering vehicle to allow DLA to purchase nickel niobium alloy as needed within the parameters established.

C. SCOPE

The contractor will provide the necessary management, services, personnel, and documentation required to accomplish the tasks and provide the materials described within this SOW.

To reduce the risk of unavailability in the nickel niobium supply chain, DLA Strategic Materials intends to purchase nickel niobium alloy with maximum values shown in Table 1 over a five-year ordering period.

Table 1 - Maximum Quantities (in lbs)

Execution Year	Nickel-Niobium Alloy (lbs)
Year One	16,132.28
Year Two	401,997.25
Year Three	387,011.79
Year Four	380,836.3
Year Five	260,136.65
TOTAL	1,446,114.27

D. COMPLIANCE REQUIREMENTS

The contractor must meet the below listed compliance requirements:

- Harmonized Tariff Schedule, the Tariff Act of 1930, and the Trade Act of 1974
- United States (US) Customs and Border Protection (CBP) Form 7501
- US CBP Form 6059 General Declaration
- US CBP Form 5106 Request for Importer Number or Notification of Importer Number

II. PERFORMANCE REQUIREMENTS

This procurement will be unrestricted under North American Industry Classification System (NAICS) 331410 **Nonferrous Metal (except Aluminum) Smelting and Refining** with a size standard of 1,000 employees.

Unless otherwise specified, all references to days are considered business days. The contractor will perform actions as necessary to accomplish the objectives stated above as well as the following tasks and deliverables.

A. MATERIAL REQUIREMENTS

1. Nickel niobium alloy

- The material must be nickel niobium alloy consistent with the requirements identified below. All testing methods must be reported along with the components to which they were used to test.

Table 2 - Nickel Niobium Alloy specifications

Component	Limit		Unit
Nb	range	58-72	%
Al	max.	1.5	%
Fe	max.	0.6	%
O	max.	0.1	%
N	max.	0.03	%
Ag	max.	10	ppm
As	max.	50	ppm
B	max.	20	ppm
Bi	max.	5	ppm
C	max.	500	ppm
Co	max.	500	ppm
H	max.	50	ppm
Ni	remainder		
P	max.	200	ppm
Pb	max.	50	ppm
S	max.	150	ppm
Sb	max.	50	ppm
Se	max.	10	ppm
Si	max.	4000	ppm
Sn	max.	30	ppm

Ta	max.	3000	ppm
Te	max.	10	ppm
Ti	max.	150	ppm

- b. Material must be a lump or flake material with dimensions under 50 mm in any direction with a maximum of 10% by mass of material retained on the sieve.
- c. Each lot must be approximately 19,000 kg contained in 76 drums as defined in Part II, Section F of this SOW with each drum containing approximately 250 kg. Co-mingling of multiple production lots into one larger lot will not be accepted. The contractor must assign a unique lot number to each production lot.
- d. Material must be non-radioactive, non-licensable per Nuclear Regulatory Commission (NRC) regulations. It must contain less than 0.05 percent combined thorium and uranium by mass. Th+U, Th, and U percent by mass must be reported individually on the contractor's COA for each lot of material the contractor delivers to the government.

B. SAMPLING NICKEL NIOBIUM ALLOY

1. The government will witness the contractor's sampling operations for each lot of material the contractor intends to deliver to the government for compliance with the sampling plan. The government reserves the right to waive witnessing of the sampling process on a case-by-case basis.
2. The contractor's sampling operations must take place within the Continental United States (CONUS).
3. The contractor must permit the government access to activities associated with sample collection and distribution at the time the contractor is manufacturing each lot of material the contractor intends to provide to the government. The contractor's production facilities must be within the CONUS for all lots of material the contractor intends to supply the government.
4. The contractor must provide the government with written notice of the date(s) the contractor intends to begin sampling activities of lots of material it intends to provide to the government at least 20 business days prior to that date.
5. The contractor must submit a sampling plan as part of the proposal. The contractor's sampling plan must clearly describe:
 - a. The sampling process and how it produces a sample that is representative of the lot.
 - b. Cite any relevant sampling standards used for sampling as applicable.
 - c. Provide supporting documentation showing that the sampling practice produces a sample that is representative of the lot (e.g., cite a standard, industry practice, or data demonstrating the sample is representative, etc.).
6. The contractor cannot make any changes to the sampling plan without the explicit written approval of the government representative.

- a. If the contractor wishes to make changes to the approved sampling plan, the contractor must submit the revised proposed sampling plan to the government representative for approval no later than 20 business days prior to implementation of the changes. The revised sampling plan must be submitted in PDF format by email. The government representative will advise the contractor whether the revised proposed sampling plan is approved within 5 business days of receipt. All sampling must be conducted in accordance with the approved sampling plan.
 - b. Alternatively, if subsequent contractor sampling plans and processes remain unchanged from previous sampling plans, the contractor must send an email to the government stating that there have been no changes to its sampling plan and processes at least 20 business days prior to the date the contractor begins sampling activities.
- a. The contractor must collect four equivalent samples for each lot of material the contractor intends to supply to the government – one for the Government to retain, one for the Contractor, and two for the third-party laboratory (one for analysis and one for reserve). All four samples must be of the same size and form and must be representative of the chemical properties of the lot of material. All four samples must have a minimum sample size that satisfies the requirements of the independent third-party laboratory for sample size.
- 7. The contractor must ship the government's sample for each lot with each lot of material it delivers to the government.
- 8. The contractor must document and transfer chain of custody of the lot's analytical sample to the independent third-party laboratory.

C. TESTING NICKEL NIOBIUM ALLOY

- 1. The contractor must employ an independent third-party analytical chemistry laboratory (lab) to chemically characterize each lot of material the contractor intends to deliver to the government. The lab must analyze each lot in accordance with Table 2 above.
- 2. The contractor must document this information to the government in a certificate of analysis (COA). The contractor must deliver a COA for each representative sample from each lot for which chemical analysis was performed.
- 3. The third-party lab used to perform verification testing must:
 - a. Have CONUS-based operations.
 - b. Perform all analytical chemistry activities in this SOW within the CONUS.
 - c. Be certified in International Organization for Standardization (ISO) 17025 through A2LA with the following specific certifications (at a minimum): sample preparation for mineral ore, limestone, ferro-alloys (falling within the A2LA P112 flexible scope policy); inductively coupled plasma (ICP) for Al, Ag, As, B, Ba, Bi, Ca, Cd, Ce, Co, Cr, Cu, Fe, K, La, Li, Mg, Mn, Mo, Na, Nb, Ni, P, Pb, Sb, Se, Si, Sn, Sr, Ta, Ti, V, W, Zn (falling within the A2LA P112 flexible scope policy); combustion analysis for C, N, O, S (falling

within the A2LA P112 flexible scope policy); wavelength-dispersive x-ray fluorescence (WDXRF) analysis for Al, Ca, Cr, Co, Cu, Fe, Mn, Mo, Na, Nb, Ni, P, S, Si, Ti, V (falling within the A2LA P112 flexible scope policy); gravimetric precipitation analysis for CaF₂, Mo, Nb, S, SiO₂; volumetric analysis for B, Ca, Cr, Fe, Metallic Fe, Mg, Mn, Ni, Ti, V; gas evolution for CO₂; and miscellaneous gravimetric analysis for sieve testing, loss on ignition, moisture at 105°C.

- d. Have capability and experience in analytical chemistry techniques and methods and method development in wet chemistry (e.g., sample preparation and digestion, titration, reagent preparation, glovebox work, etc.), and instrumental analysis.
 - e. Have capability and experience in clean pulverization capabilities, laboratory sample preparation and digestion, titration, and reagent preparation.
 - f. Have capability and experience in instrumental analysis including ICP optical emission spectroscopy/atomic emission spectroscopy (ICP-OES/AES), ICP-Mass Spectrometry (ICP-MS), and inert gas fusion and combustion analysis specifically using LECO or similar instruments.
4. The Contactor must submit with their proposal the name of the independent third-party laboratory that they intend to use to test material intended for delivery to the government.
 5. The contractor must submit for approval the third-party laboratory's testing plan no later than 30 business days after contract award. The testing plan must be approved prior to the commencement of any sampling operations.
 6. Use of the contractor's selected third-party laboratory is conditional upon approval by the government.
 7. The testing plan must include (at a minimum):
 - a. Name and address of the third-party laboratory the contractor intends to use;
 - b. Summary description of sample preparation;
 - c. Summary description of the methodology and instrumentation (as applicable) the lab will employ for each required data value in the SOW material specifications;
 - d. Any significant assumptions the lab might elect to make that are beyond the accepted assumptions made for chemical analysis (instruments and wet chemistry); and
 - e. Acknowledgement of the safety portion of the safety data sheet (SDS) for the specific material being tested.
 8. The contractor cannot make any changes to the independent third-party analytical chemistry laboratory's testing plan (i.e., the proposed analytical methodologies the independent laboratory will employ that will be submitted as part of the contractor's proposal) without the explicit written approval of the government representative.
 - f. If the contractor or 3rd party laboratory wishes to make changes to the approved testing plan, the contractor must submit the revised proposed testing plan to the government representative for approval no later than 20

business days prior to implementation of the changes. The revised testing plan must be submitted in PDF format by email. The government representative will advise the contractor whether the revised proposed testing plan is approved within 5 business days of receipt.

- g. Alternatively, if subsequent contractor testing plans and processes remain unchanged from previous testing plans, the contractor must send an email to the government stating that there have been no changes to its testing plan and processes at least 20 business days prior to the date the contractor begins testing operations..
- 9. All testing must be accomplished in accordance with the approved testing plan. The Government will reject material as non-compliant if the contractor does not comply with the Testing Plan.
- 10. The contractor must document chain of custody of the analytical and laboratory reserve samples. The contractor must produce the chain of custody documents within 24 hours when requested by the Government.

D. CERTIFICATES OF ANALYSIS

1. The Contractor must submit a sample COA of material which meets the requirements described in Part II, Section D, paragraph 8 of this SOW. If the Contractor's submitted COA does not meet the requirements described in paragraph 8 of this section, the Contractor must briefly describe the process changes they intend to make to ensure their COA will meet the requirements of II.D.8. of the SOW.
2. The contractor must provide a COA with each production lot as defined within this SOW. Final approval of each Contractor COA takes place prior to Contractor's shipment of each lot of material to the Government.
3. The contractor must affix a cover letter with their letterhead to each COA the contractor submits to the government. The contractor's letterhead must include all information the contractor is required to report and that information is not generated by a separate entity. For example, the third-party must generate an analytical report on a lot of material and the contractor cannot alter that report in any manner but will affix a cover letter to that analytical report from the third-party lab.
4. The COA is a legal document, which attests that the materials delivered by the contractor meet the specifications as stated in this SOW. The government will determine based on the COA if the lot(s) of the materials are acceptable for delivery. The contractor must carefully sample and test the material to ensure the reliability of each submitted COA.
5. The COAs must be submitted by email in PDF format to the government for review and acceptance prior to shipping any lot(s) to the point of delivery.
6. The COA must have a statement certifying that the material meets and/or exceeds required material specifications as stated within this SOW.

7. The government POC will review every COA submitted to determine acceptability and compliance with the requirements in this SOW within 10 business days after receipt.
8. Each COA must include the following (at a minimum):
 - a. Lot number.
 - b. Weight and number of containers the COA covers.
 - c. List of tests performed and/or the methodology applied. The contractor must include any significant assumptions made other than typical assumptions for test methodology as a footnote to the reported value.
 - d. Numerical results and qualitative results (as applicable).
 - e. Error ranges/limits for results.
 - f. Specifications/requirements.
 - g. Name of the company or name of the laboratory performing the testing.
 - h. Date(s) each analysis is performed.
 - i. Printed name and signature of analyst.
 - j. Results for Uranium and Thorium content
9. The COA and all documents submitted to the government must be in English.

E. ENVIRONMENTAL, SAFETY, AND OCCUPATIONAL HEALTH

1. The government will ensure compliance with all federal, state, and local regulations as well as international laws (as applicable) for the use, transport, emergency response, and disposal of the acquired material.
2. The contractor must ensure compliance with the Department of Transportation (DOT) regulations by certifying hazardous materials are properly classified, described, packaged, marked, labeled, and are in a safe condition to transport the material (49 CFR 173).
3. The contractor must ensure compliance with the Occupational Safety and Health Administration (OSHA) regulations regarding the proper management, handling, storage, transportation, and disposition of material (29 CFR 1910).
4. The contractor must communicate the hazards of the material by providing the SDS in accordance with The United Nations Globally Harmonized System of Classification and Labeling of Chemicals (GHS) (29 CFR 1910).
5. The contractor must ensure compliance with the Environmental Protection Agency (EPA) regulations for the characterizing of waste to determine appropriate disposal of material and any associated packing and shipping material (40 CFR 260).
6. The contractor must ensure compliance with Strategic Materials Environmental, Safety, and Occupational Health (ESOH) Depot Orientation Guide and ESOH Policy Statement available on the Strategic Materials website:
<https://www.dla.mil/Strategic-Materials/>.

F. PACKAGING

1. Nickel niobium alloy.

- a. Material must be packaged in new UN 1A2 compliant steel drums with a rated capacity of 55 gallons or metric equivalent capacity; no more than four drums per pallet.
 - b. The contractor must package all material delivered throughout the performance of this SOW in the same size and type of drum (identical in all aspects).
 - c. The contractor must submit detailed design drawings and manufacturer's certifications for all proposed steel drums for use under this SOW for review and approval from the government QAS prior to their utilization.
 - d. No steel drums can be utilized that have not been preapproved for use under this SOW by the government.
 - e. The steel drums must be new. Use of used, recycled, repaired, remanufactured, or refurbished drums is not allowed.
 - i. The steel drums must have a bolt and ring closure.
 - ii. The steel drums must have a gasket made of a synthetic elastomer to prevent the admission of moisture and to prevent leakage of its contents, especially fine particulates. The gasket's material must have a chemical resistance consistent with the contents of the drum.
 - iii. Both the steel drums' exteriors and interiors must be hot dipped, galvanized, or painted with a rust preventive coating.
 - iv. A copy of the drum manufacturer's certification must be packaged with each shipment representing all drums in that shipment.
2. Upon delivery the drums must be intact and free of damage, contamination, and leaks.
 3. Palletizing requirements. All container/items must be placed onto new undamaged pallets meeting the specifications listed below and be banded together to prevent container/item movement during handling, transportation, or other operations where containers/items and/or personnel could be damaged and/or injured if the container/item shifts. The use of used, recycled, repaired, remanufactured, and/or refurbished pallets is not allowed.
 - a. Pallet design.
 - i. All pallets must be 48" deep (front to back) and 48" wide (side to side) in size.
 - ii. All pallets must contain four full-length solid stringers. One stringer must be located at each edge (side), and two stringers must be located in the interior of all pallets. Interior stringers must be spaced 18.5" (on center) from the outside of each edge (side). Stringers must be 1 5/8" in width and 3.5" in height. Segmented, composite, and/or block stringers are not allowable.
 - iii. All pallets must have flush sides (e.g., no wings).
 - iv. All pallets must contain "top decking" and "bottom decking" boards. All decking boards must be 6" in width with no more than 1" between boards. All decking boards must be 3/4" in thickness and be solid.

- v. All pallets must have a “two-way entry” configuration.
 - vi. Pallets may be reversible in design.
 - vii. All pallets must have a minimum rated load capacity of 750 pounds per square foot.
- b. Pallet construction.
- i. All pallets must satisfy the requirements listed in ASTM D6199-18a.
 - ii. All pallets must be constructed entirely (except fasteners) of high-density hardwood meeting the requirements of paragraph 6 of ASTM D6199-18a. Only high-density hardwood species categorized in “Group IV” of ASTM D6199-18a are allowable for pallet construction. The only allowable high density hardwood species used for pallet construction are beech, birch, hackberry, hard maple, hickory, oak (eastern), pecan, rock elm, American elm, white oak, and green oak. Pallets may be constructed by mixing together or using different high density hardwood species listed above interchangeably.
 - iii. All pallets must satisfy the requirements listed in International Standards for Phytosanitary Measures (ISPM) 15 (e.g., heat treated etc.).
 - iv. Only driven nails that are “helically” threaded, “annularly” threaded or “fluted” are allowable for use in pallet construction. The fastener length must be sufficient to provide a minimum penetration of 1.25” into the stringer.
4. An SDS specific for each grade of the contained material must be accessible and external to the packed material with each delivery. All packages must meet all requirements presented in Part II, Section E of this SOW to ensure compliance with federal, state, and local regulations for the transporting, handling, and management of material.

G. LABELING

1. All individual containers or ingot pallets must be identified with an attached tags(s)/label(s) as follows:
 - a. The font size must be of sufficient size to be legible to the unaided eye.
 - b. All information on tag(s)/label(s) must be in English.
 - c. Any ink used to print tag(s)/label(s) must be indelible, permanent, and suitable for use in unheated warehouses where temperature extremes (both hot and cold) exist. The use of paint on tags/labels to enter the required information on each tag/label is not allowable.
 - d. The tag/label material must be durable and suitable for use in unheated warehouses where temperature extremes (both hot and cold) exist. The tag/label material must not react with the commodity being tagged/labeled.
 - e. Tag/label material must have an expected lifespan of a minimum of 20 years with no degradation.

- f. All tag(s)/label(s) must be of sufficient size to permit the listing of all required information.
2. The information below must be listed on each tag/label or included with the ingot. The marking must not include a security classification or anything indicating DLA Strategic Materials ownership other than the contract and delivery order number.
 - a. Commodity name
 - b. Commodity grade, type, or form
 - c. Merge number
 - d. Country of origin
 - e. Producer name
 - f. Producer's part number
 - g. Gross weight in pounds and metric units (metric units to be in parentheses)
 - h. Net weight in pounds and metric units (metric units to be in parentheses)
 - i. Lot number
 - j. Ingot serial number. The serial number is the individual ingot number in relation to the total number of ingots in the lot (e.g., ingot 25 of 100 ingots in the lot). Each ingot will be stamped on opposite ends with that ingot's serial number corresponding to that ingot's particular lot.
 - k. Contract and delivery order number.
3. A separate metal tag must be attached to each individual container or ingot with a laser etched 3d barcode containing data listed in paragraph 2 of this section (II.G.2.).
4. Tag(s)/label(s) must be attached to each container in a permanent manner that prevents accidental detachment. The means of attachment must be suitable for use in unheated warehouses where temperature extremes (both hot and cold) exist.
5. For non-ingot material, one tag/label must be attached to the exterior of each drum container in a permanent manner that prevents accidental detachment. A second tag/label must be attached to the topside of the lid of each drum. A third tag/label must be attached to the underside of the lid of each drum. The means of attachment must be suitable for use in unheated warehouses where temperature extremes (both hot and cold) exist. Attachment by adhesive means (e.g., glue, adhesive backed tag/label etc.) is acceptable.
6. All packages must meet all requirements presented in Part II, Section E of this SOW to ensure compliance with federal, state, and local regulations for the transporting, handling, and management of material.

H. TRANSPORTATION

The contractor is responsible for transport as well as for carrier and insurance paid to (CIP) of the material to the point of delivery at one of the government storage depots listed below as indicated on each delivery order.

I. DELIVERY

1. The Contractor must deliver materials to the one of the following Government Storage Depots (delivery point) to be specified on each delivery order:

- a. Defense Logistics Agency Strategic Materials Scotia Depot
Route 5, Building #12
Scotia, NY 12302-9463
Telephone: (518) 370-3347

Unloading Hours: 7:30 AM – 2:30 PM (EST), Monday – Friday

Accessibility: Truck

Deliveries will only be accepted between the hours of 7:30 AM – 2:30 PM (EST), Monday – Friday (excluding federal holidays).

- b. Defense Logistics Agency Hawthorne Army Depot
1 South Main Ave.
Hawthorne, NV 89415-9404
Telephone: (775)-945-7001

Shipping Hours: 0800 to 1430 Pacific Standard Time (PST) Monday through Thursday. (Exceptions of Federal, State and local holidays and days of closure due to inclement weather and/or other emergencies.)

Accessibility: Truck/Rail* scale is available (Truck scale weight capacity is 90,000 lbs. and the rail scale capacity is 500,000 lbs.).

*Note: Accessibility by rail is currently unavailable but may become available in the future.

Truck Lot Receiving Procedures: Transportation companies are required to use the (CAS) Carrier Appointment System SDDC for deliveries to Hawthorne Army Depot. Transportation companies must schedule an appointment 24 hours in advance by CAS or by calling 775-945-7950 or 775-945-7949. Hawthorne Army Depot must review passports, naturalization papers and citizenship for admittance and reserves the right to reject non-compliant transportation. Any deliveries must be pre-approved prior to arriving in Hawthorne.

- c. Defense Logistics Agency Strategic Materials Hammond Depot
3200 Sheffield Avenue
Hammond, Indiana 46327-1003
Telephone: (219)-937-5383

Shipping/Receiving hours: 7:00 am to 2:30 pm (Central Time) Monday through Friday (exceptions of federal, state, and local holidays and days of closure due to inclement weather and/or other emergencies)

Accessibility: Truck and rail.

- d. Defense Logistics Agency – Columbus
3990 E. Broad St.
Columbus, OH 43218

2. The Contractor must submit written documentation that identifies the proposed material delivery schedule for the duration of the contract to the Government within 30 business days of contract award. This material delivery schedule shall include proposed individual delivery dates and the total net weight of each proposed delivery.
3. The contractor must schedule each delivery with the government representative only after the COA is accepted.
4. The contractor must schedule in writing with the government representative all deliveries a minimum of 15 business days in advance of delivery and receive written acceptance of the proposed delivery schedule from the government representative prior to any shipping. The notification must include the following information:
 - a. Total net weight of material to be delivered.
 - b. Total net weight of each lot of material to be delivered.
 - c. Total net weight of material contained in each package.
 - d. Total number of sub-packaging units contained inside each package if applicable.
 - e. Total number of packages
 - f. Quantity of packages on each pallet.
 - g. Pallet size (e.g. 48" x 48")
 - h. Total number of truck loads to be delivered each month.
 - i. Total number of lots contained in each truckload
 - j. Type of truck planned for transporting the material to the delivery point (e.g., enclosed van truck, overseas container etc.)
5. The contractor must take no longer than 12 months to deliver material once a delivery order is issued.

J. INSPECTION AND ACCEPTANCE

1. DLA personnel will perform in-depth visual inspections of material upon transferring to DLA custody at the final delivery point to verify conformance with the requirements stated in the SOW that can be verified visually.
2. The government reserves the right to sample and test each lot of material received to verify the material meets the specifications set forth in this SOW.

3. The government representative will provide written notification of the acceptance/rejection of each delivered lot of material to the contractor within 30 business days of delivery. The contractor cannot invoice until the government representative has issued a written notification of acceptance of each lot of material delivered.
4. If any delivered lot of material or other deliverable does not conform to the requirements stated in this SOW, the government reserves the right to reject the non-conforming material.
5. Material rejected based on non-compliance must be replaced at no cost to the government. The contractor is responsible for removal and replacement of rejected lots of material, including all attendant costs within 90 calendar days of the final delivery.